MARKING SCHEME

Model Paper — 04

Mathematics II

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Note (for teachers):

- Candidates answer **five** questions from Part A (Questions 1 – 6) and **five** questions from Part B (Questions 7 – 12). Each question carries 10 marks. Total 100 marks.
- The marks for each step are shown in the right-hand column and add up to the marks printed beside each part of the question paper.
- Award full marks for any correct alternative method. Accept answers within the stated ranges for values read from graphs, tables or measured from constructions. Penalise a missing or wrong unit only once in a question.

Part A — Answers

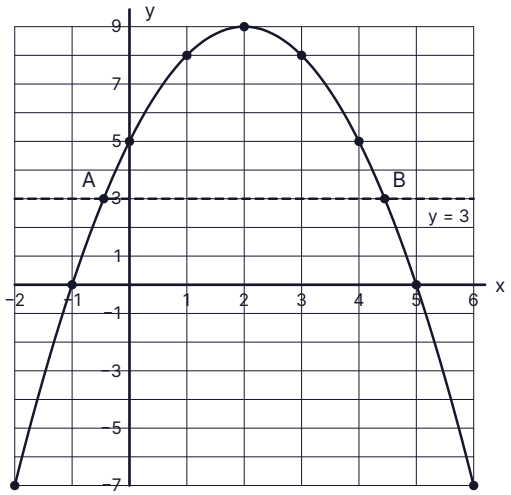
Question 1

(i)	
Interest for 6 months = Rs 10 080, so interest for 1 year = 10 080 × 2 = Rs 20 160	**1**
Rate = $\frac{20\,160}{96\,000} \times 100$ = **21%**	**1**
(ii)(a)	
Capital repaid each month = 96 000 ÷ 6 = Rs 16 000	**1**
Balances at the beginning of the months: 96 000, 80 000, 64 000, 48 000, 32 000, 16 000	**1**
Sum = Rs 336 000 (rupee-months)	**1**
(ii)(b)	
Interest = sum of balances × $\frac{r}{100} \times \frac{1}{12}$, so 336 000 × $\frac{r}{100} \times \frac{1}{12}$ = 10 080	**1**
r = $\frac{10\,080 \times 12 \times 100}{336\,000}$	**1**
Annual rate of Method 2 = **36%**	**1**
(iii)	
36% > 21%, so Method 2 has the higher annual rate	**1**
Reason: in Method 2 the amount owed falls each month, so the same total interest is charged on a much smaller average balance (Rs 56 000 on average, compared with Rs 96 000 in Method 1)	**1**

Total marks: 10

Question 2

(i)	
By symmetry about x = 2, the value at x = 3 equals the value at x = 1: y = **8**	**1**
(ii)	
Suitable scale on both axes (e.g. 2 cm = 1 unit on x, 1 cm = 1 unit on y), axes labelled	**1**
Points (−2, −7), (−1, 0), (0, 5), (1, 8), (2, 9), (3, 8), (4, 5), (5, 0), (6, −7) plotted correctly (at least 7 correct)	**1**
Smooth curve through the points: an inverted parabola (∩-shaped) with maximum point (2, 9)	**1**
(iii)	
(a) x = −1 and x = 5	**1**
(b) y = −(x + 1)(x − 5) (p = 1, q = −5)	**1**
(iv)	
(a) Line y = 3 drawn; A and B read as x ≈ −0.4 and x ≈ 4.4 (accept −0.5 to −0.4 and 4.4 to 4.5)	**1**
Range of x for y > 3: **−0.4 < x < 4.4**	**1**
(b) AB = 4.4 − (−0.4)	**1**
AB ≈ **4.9** (accept 4.8 – 5.0); exact value $2\sqrt{6} \approx 4.9$	**1**



Graph of y = −(x + 1)(x − 5) with the line y = 3

Total marks: 10

Question 3

(i)	
In right-angled triangle ABC, BC = AC sin 20°	**1**
sin 20° = 0.3420 (tables)	**1**
BC = 60 × 0.3420 = **20.52 m**	**1**
(ii)	
BD = 2 BC (since DC = BC) = 2 × 20.52 = **41.04 m**	**1**
tan ∠DEB = $\frac{BD}{BE}$	**1**
= $\frac{41.04}{40}$ = **1.0260**	**1**
(iii)	
Tables: tan 45°42′ = 1.0247, with mean difference for 2′ = 0.0012 (tan 45°48′ = 1.0283)	**1**
∠DEB = **45°44′** (accept 45°42′ – 45°48′)	**1**
(iv)	
∠DEB − ∠CAB = 45°44′ − 20° = 25°44′	**1**
25°44′ > 25°, so ∠DEB exceeds ∠CAB by more than 25°	**1**

Total marks: 10

Question 4

(i)	
Area of the triangle = $\frac{1}{2} \times 2x \times 2x = 2x^2$	**1**
Area of the rectangle = $(x + 1)(x - 5) = x^2 - 4x - 5$	**1**
$2x^2 - (x^2 - 4x - 5) = 100$	**1**
$x^2 + 4x + 5 = 100$, so $x^2 + 4x - 95 = 0$ (shown)	**1**
(ii)	
$x = \frac{-4 \pm \sqrt{(16 + 380)}}{2} = \frac{-4 \pm \sqrt{396}}{2}$	**1**
$\sqrt{396} = 6\sqrt{11} = 6 \times 3.32 = 19.92$; $x = \frac{-4 + 19.92}{2} = 7.96$ (the negative root is rejected since $x > 5$)	**1**
$x \approx$ **8 cm**	**1**
(iii)	
Longest edge = $2\sqrt{2}\,x = 2 \times 1.41 \times 8$	**1**
= 22.56 cm	**1**
22.56 > 22, so the longest edge exceeds 22 cm	**1**

Total marks: 10

Question 5

(i)	
$x + y = 120$	**1**
$85x + 130y = 12\,450$	**1**
$85x + 85y = 10\,200$; subtracting: $45y = 2\,250$	**1**
$y = 50$ (large baskets)	**1**
$x = 120 - 50 = 70$ (small baskets). Check: $85 \times 70 + 130 \times 50 = 5\,950 + 6\,500 = 12\,450$	**1**
(ii)	
Let a small basket hold n kg; a large basket holds $(n + 4)$ kg	**1**
$70n + 50(n + 4) = 920$	**1**
$120n + 200 = 920$	**1**
$120n = 720$	**1**
$n = 6$; a small basket holds **6 kg**	**1**

Total marks: 10

Question 6

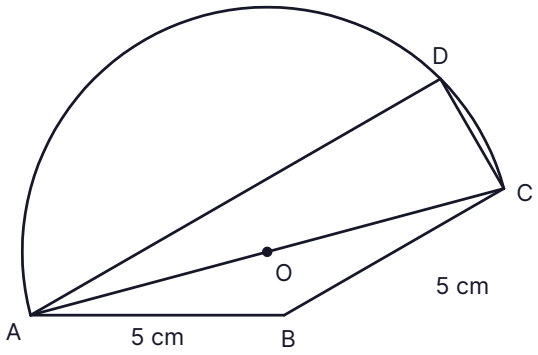
(i)	
Class mid-values: 44.5, 54.5, 64.5, 74.5, 84.5, 94.5	**1**
$\Sigma fx = 89 + 381.5 + 645 + 894 + 507 + 283.5 = 2\,800$; $\Sigma f = 40$	**1**
Mean = $\frac{2\,800}{40}$ = **70 loaves**	**1**
(ii)	
Expected total = 70 × 26	**1**
= **1 820 loaves**	**1**
(iii)	
Profit per loaf = Rs 14	**1**
Expected profit = 1 820 × 14	**1**
= **Rs 25 480**	**1**
(iv)	
Days with fewer than 60 loaves: 2 days (40 – 49) and 7 days (50 – 59); the smallest possible sales are 40 and 50 loaves per day	**1**
Minimum total = 2 × 40 + 7 × 50 = 80 + 350 = **430 loaves**	**1**

Total marks: 10

Part B — Answers

Question 7

(i)	
AB = 5 cm drawn accurately (±1 mm)	**1**
90° at B constructed (arcs and perpendicular line visible) and 60° constructed from BA (equilateral-type arcs visible) on the same side, so that ∠ABX = 90° + 60° = 150° (accept 148° – 152° if measured)	**2**
(ii)	
C marked on BX with BC = 5 cm (compass arc from B with radius AB), AC joined; AC ≈ 9.7 cm (9.5 – 9.9 accepted)	**1**
(iii)	
Arcs of equal radius (more than half of AC) drawn from A and from C, meeting on both sides of AC	**1**
Perpendicular bisector drawn; it cuts AC at its midpoint O	**1**
Semicircle with centre O and radius OA ≈ 4.8 cm drawn on the side of AC opposite to B	**1**
(iv)	
Line through A parallel to BC constructed (equal-angle construction arcs visible)	**1**
It meets the semicircle at D (AD ≈ 9.3 cm); CD joined (CD ≈ 2.5 cm)	**1**
(v)	
∠BAC = ∠BCA = (180° − 150°) ÷ 2 = 15°; AD ∥ BC so ∠DAC = ∠ACB = 15° (alternate); ∠ADC = 90° (angle in a semicircle); ∠ACD = 180° − 90° − 15° = **75°** (accept 74° – 76° if measured)	**1**



Model construction drawn to scale (construction arcs omitted)

Total marks: 10

Question 8

(i)	
DE = EC (E is the midpoint of CD)	**1**
∠DEG = ∠CEF (vertically opposite angles)	**1**
∠EDG = ∠ECF (alternate angles, AG ∥ BC); therefore △DGE ≅ △CFE (angle, side, angle)	**1**
(ii)	
From (i), EG = EF; also DE = EC. So the diagonals DC and GF of the quadrilateral DGCF bisect each other	**1**
Therefore DGCF is a parallelogram. (Alternatively DG = CF and DG ∥ CF)	**1**
(iii)	
∠BHF = ∠DEG (corresponding angles, AH ∥ DC)	**1**
∠BFH = ∠DGE (corresponding angles, BC ∥ AG); therefore the triangles are equiangular (the third angles are equal as well)	**1**
(iv)	
DG = CF (from (i)) = $\frac{4}{5}$ BC and BF = $\frac{1}{5}$ BC, so $\frac{BF}{DG} = \frac{1}{4}$	**1**
From the similar triangles, $\frac{BH}{DE} = \frac{BF}{DG} = \frac{1}{4}$	**1**
DE = $\frac{1}{2}$ DC = $\frac{1}{2}$ AB, so BH = $\frac{1}{4} \times \frac{1}{2}$ AB = $\frac{1}{8}$ AB	**1**

Total marks: 10

Question 9

(i)	
Volume of one pyramid = $\frac{1}{3} \times a^2 \times 3a = a^3$	**1**
Volume of 15 pyramids = $15a^3$	**1**
Volume of the cylinder = $\pi r^2 \times 5r = 5\pi r^3$	**1**
$15a^3 = 5\pi r^3$, so $a^3 = \frac{\pi}{3} r^3$ (shown)	**1**
(ii)	
$a^3 = \frac{3.14}{3} \times 7.6^3$; $\lg a^3 = \lg 3.14 + 3 \lg 7.6 - \lg 3 = 0.4969 + 3 \times 0.8808 - 0.4771$	**1**
$\lg a^3 = 2.6622$, so $a^3 =$ **459.5** (accept 459 – 460)	**1**
$\lg a = 2.6622 \div 3 = 0.8874$, $a = 7.71$; $a \approx$ **8 cm**	**1**
(iii)	
Volume of the cuboid = $a \times a \times 3a = 3a^3 = 3 \times 512 = 1\,536\ cm^3$	**1**
Volume of the pyramid cut out = $a^3 = 512\ cm^3$	**1**
Volume remaining = 1 536 − 512 = **1 024 cm^3**	**1**

Total marks: 10

Question 10

(i)	
30 m, 34 m, 38 m	**1**
(ii)	
$T_n = a + (n - 1)d = 30 + 11 \times 4$	**1**
= **74 m**	**1**
(iii)	
$S_{12} = \frac{12}{2}(30 + 74)$	**1**
$= 6 \times 104 =$ **624 m**	**1**
(iv)	
$S_{13} = \frac{13}{2}(2a + 12 \times 6) = 624$	**1**
$2a + 72 = 96$	**1**
$a =$ **12 m**	**1**
(v)	
$30 + 4(n - 1) = 12 + 6(n - 1)$	**1**
$18 = 2(n - 1)$, $n = 10$: on the **10th day** (both lay 66 m)	**1**

Total marks: 10

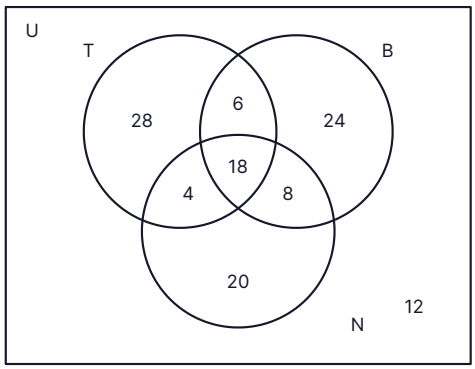
Question 11

(i)(a)	
$\angle OAT = 90°$ (the tangent is perpendicular to the radius at the point of contact)	**1**
$\angle OBT = 90°$ (same reason)	**1**
$\angle OAT + \angle OBT = 180°$ (opposite angles supplementary), so OATB is a cyclic quadrilateral	**1**
(i)(b)	
In $\triangle OAT$ and $\triangle OBT$: OA = OB (radii), OT common, $\angle OAT = \angle OBT = 90°$ (right angle, hypotenuse, side), so $\triangle OAT \cong \triangle OBT$	**1**
Hence $\angle AOT = \angle BOT$, i.e. OT bisects $\angle AOB$	**1**
(ii)	
$\angle TBA = \angle TAB$ (TA = TB, tangents from an external point, so $\triangle TAB$ is isosceles)	**2**
$\angle ACB = \angle TAB$ (alternate segment theorem: the angle between the tangent TA and the chord AB equals the angle in the alternate segment)	**2**
In $\triangle TAB$: $\angle ATB = 180° - 2\angle TAB = 180° - 2\angle ACB$, so $\angle ATB + 2\angle ACB = 180°$	**1**

Total marks: 10

Question 12

(i)	
Bought at least one item = 120 − 12 = 108	**1**
Two or more kinds = 36	**1**
Exactly one kind = 108 − 36 = **72**	**1**
(ii)	
Let the number who bought all three be x; then the number who bought exactly two kinds is also x	**1**
x + x = 36	**1**
x = **18**	**1**
(iii)	
Only tea and buns = 6; exactly two kinds = 18, so the other two regions total 12. Let only T and N = y; only B and N = 2y: 3y = 12	**1**
Only tea and newspapers = **4**; only buns and newspapers = **8**	**1**
(iv)	
n(T) = 56: only T = 56 − (6 + 4 + 18) = 28; n(N) = 50: only N = 50 − (4 + 8 + 18) = 20	**1**
Only buns = 72 − 28 − 20 = **24**	**1**



Completed Venn diagram (12 outside the circles)

Total marks: 10